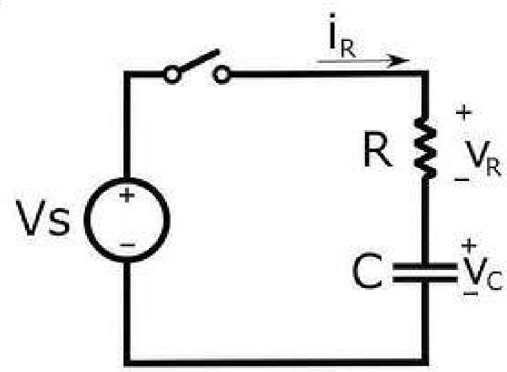
**Assignment ONE**

**SPH 2172**

**ATTEMPT all the Question individually (30 Marks)**

**To be handwritten and submitted on or before 10th April, 2026**

1. Consider a simple pendulum, having a bob attached to a string that oscillates under the action of the force of gravity. Suppose that the period of oscillation of the simple pendulum depends on its length (l), mass of the bob (m) and acceleration due to gravity (g). Derive the expression for its time period using method of dimensions (5 Marks).
2. Use Gauss theorem to derive an expression for the electric field at a point due to a thin infinitely long straight line of charge of uniform charge density ? (5 Marks)
3. Derive the expression of charging and discharging of a capacitor for both the current and voltage. Illustrate these expressions graphically (15 Marks).




4. Derive the expression of stored charge in a capacitor (3 Marks).
5. A capacitor of C=100 μF is connected in series with a resistor of R=10kΩ to a 12 V battery. Determine the time constant, Voltage across capacitor at t=1s and the charge at that time
   i. Determine the charge stored at 0.5 s
   ii. Determine the current stored at 0.5 s (7 Marks)

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